Regression, Correlation And Hypothesis Testing

Chapter Overview

1: Exponential Models

Recap of Pure Year 1. Using $y = ab^x$ to model an exponential relationship between two variables.

2: Measuring Correlation

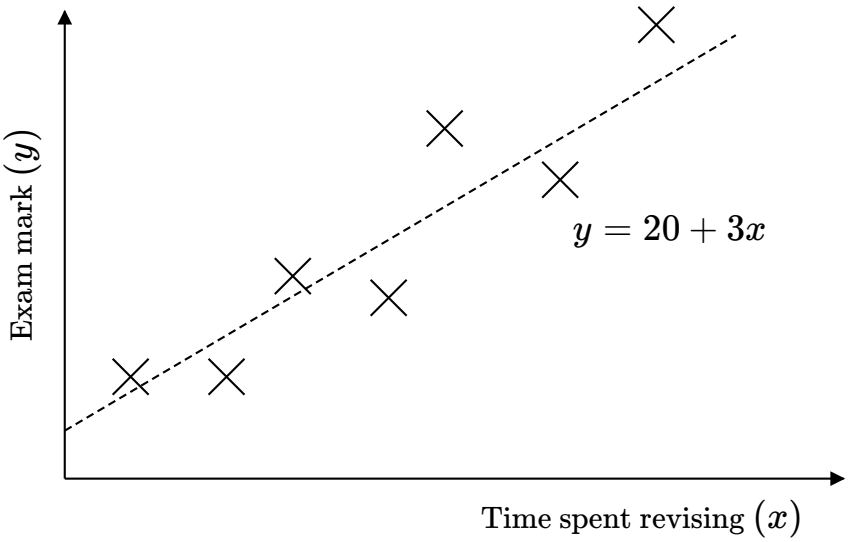
Using the Product Moment Correlation Coefficient (PMCC), r, to measure the strength of correlation between two variables.

3: Hypothesis Testing for no correlation

We want to test whether two variables have some kind of correlation, or whether any correlation observed just happened by chance.

RECAP: What is regression?

Solution 3

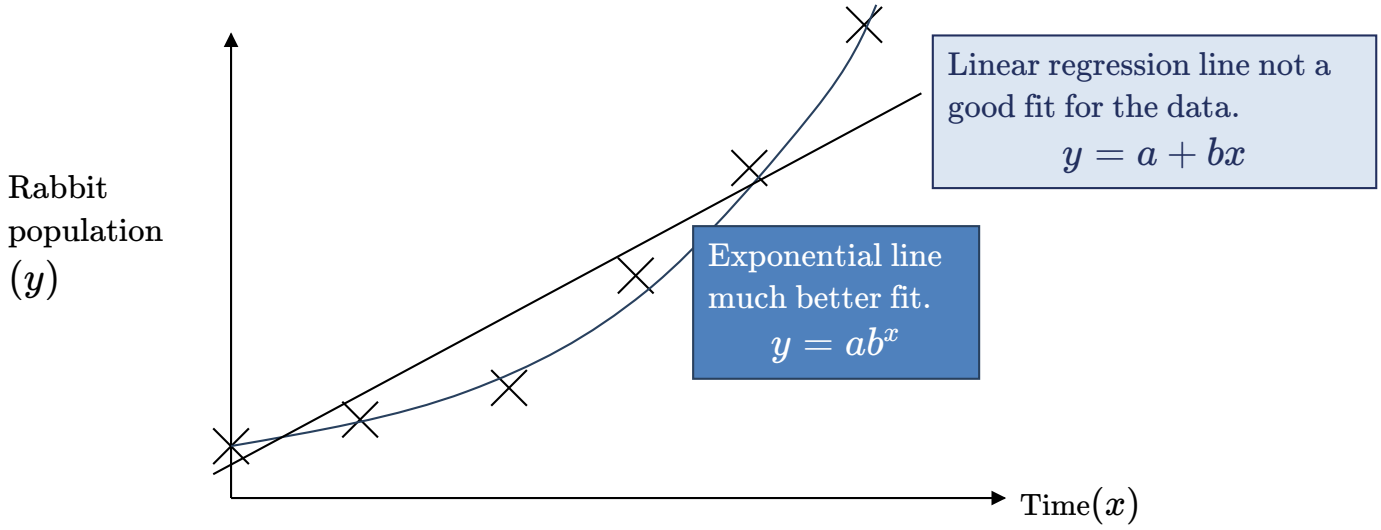


'Regression' is the act of setting the parameters of our model to best explain the data
(here the parameters are the gradient and y-intercept of the line of best fit)

I record people's exam marks as well as the time they spent revising. I want to predict how well someone will do based on the time they spent revising. How would I do this?

RECAP: Exponential Regression

Solution 4



For some variables, e.g. population with time, it may be more appropriate to use an **exponential** equation, i.e. $y = ab^x$, where a and b are constants we need to fix to best match the data.

$$y = ab^x$$

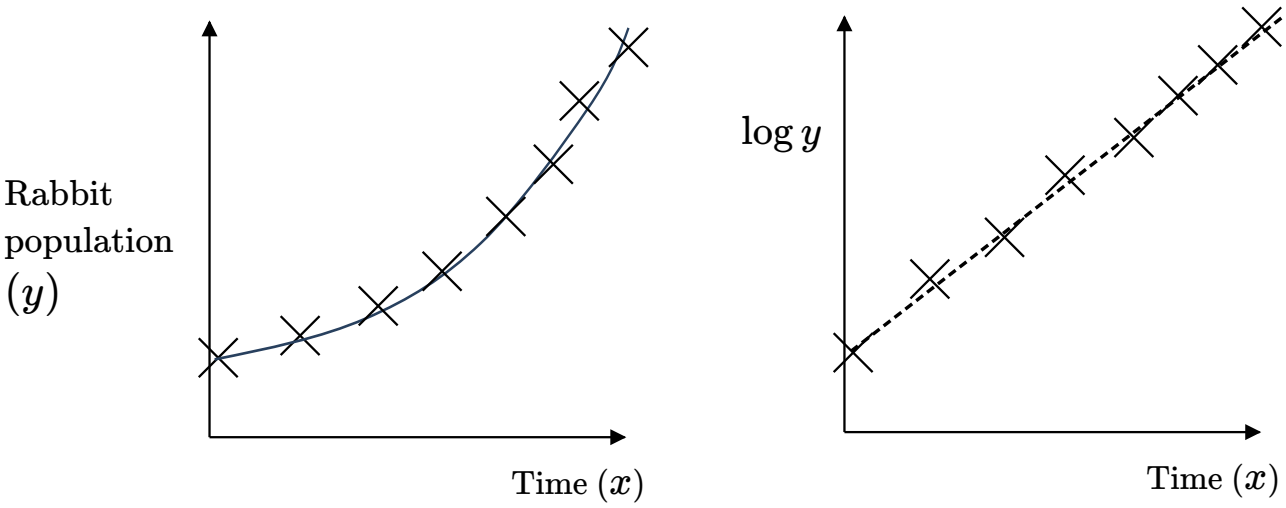
$$\log y = \log(ab^x)$$
$$\log y = \log a + x \log b$$

If $y = kb^x$ for constants k and b then $\log y = \log k + x \log b$

RECAP: Exponential Regression

Solution 5

If $y = kb^x$ for constants k and b then $\log y = \log k + x \log b$



Comparing the equations, we can see that if we log the y values (although leave the x values), the data then forms a straight line, with y-intercept $\log k$ and gradient $\log b$.

Modelling growth rate against temperature

Solution 6

[Textbook] The table shows some data collected on the temperature, in $^\circ$C, of a colony of bacteria (t) and its growth rate (g).

Temperature, $\mathbf{t}$ ($^\circ$C)	3	5	6	8	9	11
Growth rate, g	1.04	1.49	1.79	2.58	3.1	4.46

The data are coded using the changes of variable $x = t$ and $y = \log g$. The regression line of y on x is found to be $y = -0.2215 + 0.0792x$.

a. Mika says that the constant -0.2215 in the regression line means that the colony is shrinking when the temperature is 0°C. Explain why Mika is wrong

b. Given that the data can be modelled by an equation of the form $g = kb^t$ where k and b are constants, find the values of k and b.

a

a

At $t = 0, y = -0.2215$ but $g = 10^{-0.2215} = 0.600$ (3 s.f.). The growth rate is positive.

b

b

$\log g = \log k + t \log b = -0.2215 + 0.0792t$.

$k = 10^{-0.2215} = 0.600, b = 10^{0.0792} = 1.20$ (3 s.f.).

The textbooks starts with $y = -0.2215 + 0.0792x$ and raises 10 to the power of each side. You can also start with $g = kb^t$ and then log.

Test your Understanding — Solution 7

Robert wants to model a rabbit population P with respect to time in years t. He proposes that the population can be modelled using an exponential model: $P = kb^t$

The data is coded using $x = t$ and $y = \log P$. The regression line of y on x is found to be $y = 2 + 0.3x$. Determine the values of k and b.

$P = kb^t$

$\log P = \log k + t \log b$

$y = \log k + x \log b$

$\therefore \log k = 2 \quad \rightarrow \quad k = 100$

$\log b = 0.3 \quad \rightarrow \quad b = 2.00 (3sf)$

Geogebra Regression model

Power regression on the CG50 — Solution 8

A website has had the following annual numbers of page views each year:

Year	2012	2013	2014	2015	2016	2017	2018	2019
Views **V**	10115	26790	60306	180386	1119801	8.3 m	21.9 m	*57.5 m*

Determine the appropriate constants if we assume a power model $V = at^b$, where t is the number of years after 2011.

CG50 instructions

1. *Select statistics mode*
2. *Enter the list (e.g. the ones above where year starts from 1)*
3. *Press the Calc Menu (F2)*
4. *Press the REG menu (F3)*
5. *There are a variety of regressions. Go to Next Page (F6)*
6. *Select Power for power regression*

$V = at^b$

$\log V = \log a + b \log t$

So need to log the t values and V values.

logt	0	0.3010	0.4771	0.6020	0.6989	0.7781	0.8450	0.9030
logV	4.0050	4.4280	4.7804	5.2562	6.0491	6.9191	7.3404	7.7597

Using calculator stats mode, y-intercept: **loga = 3.3406**, gradient: **b = 4.3025**

Therefore $V = 2191 \times t^{4.3025}$

Fitting a power model to page views

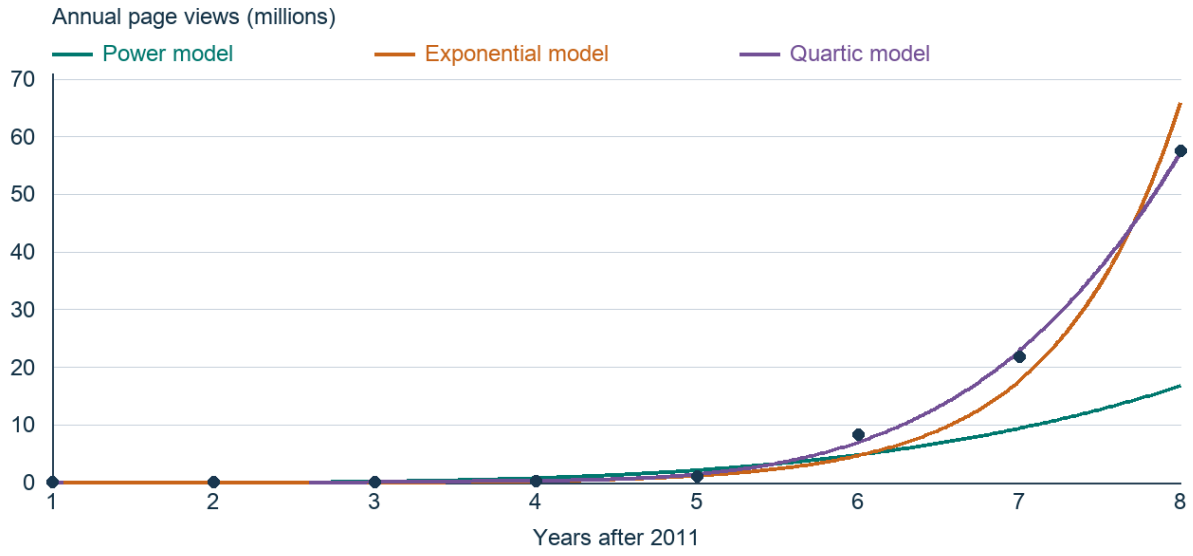
Solution 9

A website has had the following annual numbers of page views each year:

Year	2012	2013	2014	2015	2016	2017	2018	2019
Views $\mathbf{V}$	10115	26790	60306	180386	1119801	8.3 m	21.9 m	57.5 m

Determine the appropriate constants if we assume a power model $V = at^b$, where t is the number of years after 2011.

The power model underestimates the later years.
The exponential model follows the later growth more closely.

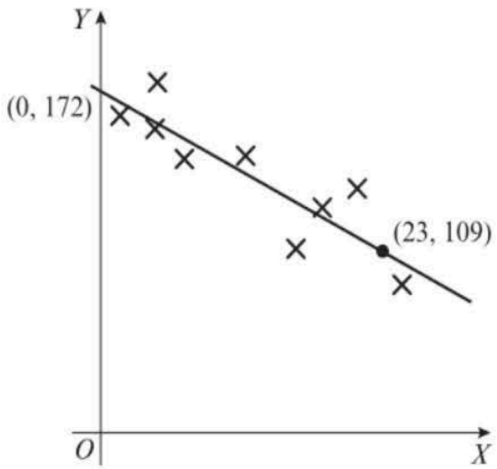


Exercise 1A – Statistics Year 2 Pages 3-5

Solution 10

3. The scatter diagram shows two coded data sets $X = \log x$ and $Y = \log y$. The regression line of Y on X passes through $(0, 172)$ and $(23, 109)$.

The original data follow a model $y = ax^n$. Find a and n to three decimal places.



3

$Y = mX + c$, so $\log y = m \log x + c$ and $c = \log a$.

The intercept is 172, giving $a = 10^{172}$.

Using $(23, 109)$, $109 = 23m + 172$.

$n = m = \frac{109-172}{23} = -\frac{63}{23} \approx -2.739$.

Exercise 1A – Statistics Year 2 Pages 3-5

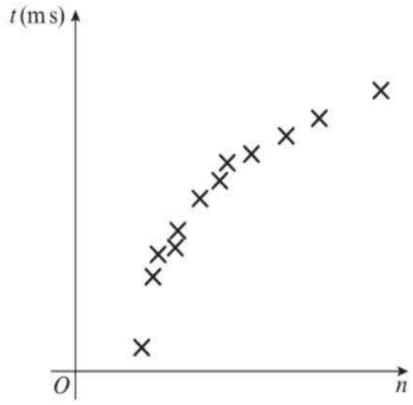
Solution 11

5. The time t ms needed for a computer algorithm to determine whether a number n is prime is recorded for different values of n. The graph shows t against n.

(a) Explain why $t = a + bn$ is unlikely to fit these data.

Using $y = \log t$ and $x = \log n$, the regression line of y on x is $y = -0.301 + 0.6x$.

(b) Find $t = an^k$, where a and k are constants.



a

A linear model gives a straight line. The points on the scatter diagram follow a curve.

b

$\log t = -0.301 + 0.6 \log n.$

$t = 10^{-0.301} \times 10^{0.6 \log n}$

$= 10^{-0.301} \times 10^{\log n^{0.6}}$

$= 10^{-0.301} n^{0.6}.$

Hence $a \approx 0.500$ and $k = 0.6$.

Exercise 1A – Statistics Year 2 Pages Page 3-5

Solution 12

6. The number of units c of a catalyst added to a chemical process and the rate of reaction r are recorded. The data are coded using $x = \log c$ and $y = \log r$. The regression line of y on x is $y = 1.31x - 0.41$.

Determine an expression for r in terms of c.

6

$\log r = 1.31 \log c - 0.41$

$r = 10^{1.31 \log c} \times 10^{-0.41}$

$= 10^{\log(c^{1.31})} \times 10^{-0.41}$

$\approx 0.389 c^{1.31}$ (3 s.f.).

7. Heights h cm and masses m kg are recorded for a sample of Galapagos penguins. With $y = \log m$ and $x = \log h$, the regression line of y on x is $y = 0.0023 + 1.8x$.

Find $m = ah^n$, giving the values of a and n.

7

$\log m = 0.0023 + 1.8 \log h$

$m = 10^{0.0023} \times 10^{1.8 \log h}$

$= 10^{0.0023} \times 10^{\log(h^{1.8})}$

$= 10^{0.0023} h^{1.8}$.

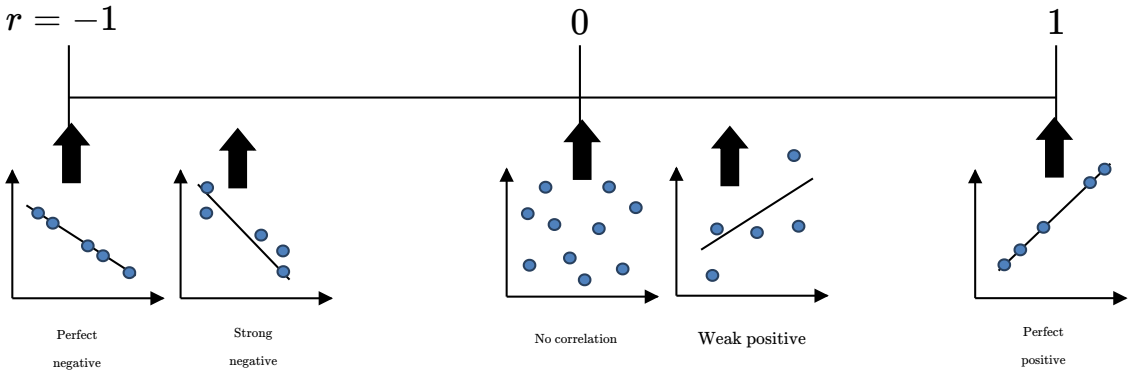
Thus $a = 10^{0.0023} \approx 1.01$ (3 s.f.) and $n = 1.8$.

Measuring Correlation

You're used to use qualitative terms such as "positive correlation" and "negative correlation" and "no correlation" to describe the **type** of correlation, and terms such as "perfect", "strong" and "weak" to describe the **strength**.

The **Product Moment Correlation Coefficient** is one way to quantify this:

The product moment correlation coefficient (PMCC), denoted by r, describes the linear correlation between two variables. It can take values between -1 and 1.



Rule of thumb: $r < -0.7$ or $r > 0.7$ is considered to be 'strong' correlation.

Note that PMCC is only applicable for a linear correlation, i.e. closeness of fit to a linear regression line (i.e. a straight 'line of best fit'). It may be the data exhibits strong correlation with respect to a different model (e.g. exponential) even when the PMCC is low.

Calculating r in the calculator

Solution 15

You must have a calculator that is capable of calculating r directly: in the A Level 2017+ syllabus you are no longer required to use formulae to calculate r.

x	y
1	3
2	6
3	5
4	8



$y = a + bx$

Data Entry

PMCC

The following instructions are for the Casio CG50.
Press MODE then select 'Statistics'.

We want to measure **linear** correlation, so select $y = a + bx$

Enter each of the x values in the table on the left.

to obtain r (i.e. the coefficients of your line of best fit and the PMCC), Press:
Graph (F1)
GRAPH1 (F1)
CALC (F1)
X (F2)
a+bx (F2)

You should obtain r = 0.868

Example

Solution 16

[Textbook] From the large data set, the daily mean windspeed, w knots, and the daily maximum gust, g knots, were recorded for the first 10 days in September in Hurn in 1987.

Day of month	1	2	3	4	5	6	7	8	9	10
w	4	4	8	7	12	12	3	4	7	10
g	13	12	19	23	33	37	10	n/a	n/a	23

a. State the meaning of n/a in the table above.
b. Calculate the product moment correlation coefficient for the remaining 8 days.
c. With reference to your answer to part b, comment on the suitability of a linear regression model for these data.

[a] a The daily maximum gust was unavailable on those days.

[b] b $r = 0.9533$.

[c] c
The correlation is strongly positive. The points lie close to a straight line, so a linear regression model is suitable.

This is a common exam question. The important bit is **evaluating the suitability of the chosen model** (in this case a linear regression model, i.e. line of best fit). The closer r is to 1 or to -1, the more suitable this linear regression model.

Exercise 1B – Statistics Year 2 Pages 6-8

Solution 17

4. The number of atoms n of a radioactive substance is measured t minutes after the start of an experiment.

t	1	2	4	5	7
n	231	41	17	7	2
$\log n$					

Use $x = t$ and $y = \log n$.

(a) Complete the table. (2 marks)

(b) Calculate the PMCC for the coded data. (1 mark)

(c) Use your answer to b to assess an exponential model. (2 marks)

The regression line is $y = 2.487 - 0.320x$.

(d) Find $n = ab^t$. (3 marks)

a

$\log n$	2.36	1.61	1.23	0.845	0.301

b $r = -0.980$ (3 s.f.).

c

The correlation between t and $\log n$ is close to -1, supporting exponential decay.

d

$\log n = 2.487 - 0.320t$.

$n = 10^{2.487}(10^{-0.320})^t$.

$a \approx 307, b \approx 0.479$ (3 s.f.).

Exercise 1B – Statistics Year 2 Pages 6-8

Solution 18

5. The widths w cm and masses m g of snowballs are recorded.

w	3	4	6	8	11
m	23	40	80	147	265
$\log w$					
$\log m$					

Code using $x = \log w$ and $y = \log m$.

(a) Complete the table. (3 marks)

(b) Calculate the PMCC for the coded data. (1 mark)

(c) Use your answer to b to assess the fit of a power model. (2 marks)

(d) Determine k and n. (3 marks)

a

$\log w$	0.4771	0.6021	0.7782	0.9031	1.041
$\log m$	1.362	1.602	1.903	2.167	2.423

b $r = 0.9996$.

c

The coded data lie close to a straight line since r is close to 1. A power model $m = kw^n$ fits well.

d

$y = 0.464 + 1.88x$.

$\log m = 0.464 + 1.88 \log w$.

$m = 10^{0.464}w^{1.88}$.

$k \approx 2.91, n = 1.88$ (3 s.f.).

Exercise 1B – Statistics Year 2 Pages 6-8

Solution 19

6. Perth, seven consecutive days in August 2015. Daily mean air temperature t °C and rainfall f mm.

t	18.0	16.4	15.3	15.0	13.7	10.2	12.0
f	3.0	13.0	4.6	32.0	28.0	63.0	22.0

© Crown Copyright Met Office

(a) Calculate the PMCC. (1 mark)

(b) Use part a to assess a linear regression model. (2 marks)

6a $r = -0.833$ (3 s.f.).

6b The moderately strong negative correlation supports a linear model.

7. Camborne, seven consecutive days in May 2015. Daily total rainfall x mm and sunshine y hours.

x	2.2	tr	1.4	4.4	tr	0.2	0.6
y	5.2	7.7	5.6	0.3	5.1	0.1	8.9

© Crown Copyright Met Office

(a) Explain 'tr'. (1 mark)

(b) Calculate the PMCC. State how you treat 'tr'. (2 marks)

(c) Assess a linear regression model using part b. (2 marks)

7a A trace is a small amount of rainfall.

7b Treat traces as zero. $r = -0.473$ (3 s.f.).

7c The negative correlation is weak. Other variables may affect the relationship, or a different model may fit better.

Extension (Challenge page 8)

Solution 20

Data recorded for two variables x and y

x	3.1	5.6	7.1	8.6	9.4	10.7
y	3.2	4.8	5.7	6.5	6.9	7.6

By calculating the product moment correlation coefficients for suitability coded values of x and y state, with reasons, whether these data are more closely modelled by a relationship of the form $y = ab^x$ or a relationship of the form $y = kx^n$, where a, b, k and n are constants.

Take logarithms.

$\log x$	0.491	0.748	0.851	0.934	0.973	1.03
$\log y$	0.505	0.681	0.756	0.813	0.839	0.881

For x and $\log y$, $r = 0.985$. For $\log x$ and $\log y$, $r = 1.00$ (3 s.f.).

The second pair has the stronger linear relationship, suggesting $y = kx^n$.

Hypothesis testing for correlation

Solution 21

Student	English	Maths
Model mean	60	70
Model SD	5	10
1	63.90	70.13
2	55.24	65.99
3	58.80	80.18
4	59.65	57.16
5	66.44	72.76
6	59.53	79.82
7	57.43	71.48
8	58.33	60.56
9	67.43	69.56
10	63.11	87.13

$r = 0.219$

Student	English	Maths
Model mean	60	70
Model SD	5	10
1	60.22	74.64
2	62.25	79.15
3	61.30	75.29
4	60.61	71.35
5	55.31	74.05
6	57.13	89.73
7	57.16	70.41
8	58.96	60.31
9	56.30	71.95
10	63.23	69.95

$r = -0.094$

A spreadsheet independently generates Maths and English marks.
What is the observed PMCC between Maths and English in this sample?

r denotes the PMCC of a **sample**.

ρ (Greek letter *rho*) is the PMCC for the **whole population**.

$\therefore$ r is the test statistic, ρ is the population parameter.

Sample correlation $r = 0.219$

What is the underlying population PMCC?

Population correlation

$\rho = 0$. The two sets of marks were generated independently. A random sample's correlation may differ from the population correlation, just as a few rolls of a fair die need not show equal frequencies.

Hypothesis testing for correlation

Solution 22

Student	English	Maths
Model mean	60	70
Model SD	5	10
1	63.90	70.13
2	55.24	65.99
3	58.80	80.18
4	59.65	57.16
5	66.44	72.76
6	59.53	79.82
7	57.43	71.48
8	58.33	60.56
9	67.43	69.56
10	63.11	87.13

$r = 0.219$

Student	English	Maths
Model mean	60	70
Model SD	5	10
1	60.22	74.64
2	62.25	79.15
3	61.30	75.29
4	60.61	71.35
5	55.31	74.05
6	57.13	89.73
7	57.16	70.41
8	58.96	60.31
9	56.30	71.95
10	63.23	69.95

$r = -0.094$

So despite that $\rho = 0$ (i.e. English/Maths marks have no underlying correlation), the observed PMCC for the sample, r, could vary, and might, purely by chance, suggest some correlation. In the first sample, this happened to be positive, in the latter, negative.

If we increased the sample size (in this case 10), we'd expect the observed PMCC to more closely match the true PMCC (0).

A natural question we might therefore ask is, if we had no knowledge about the underlying population: "**is the PMCC of $r = 0.219$ statistically significant, or are Maths/English marks not correlated, and this observed correlation emerged just by chance**?"

This sounds like a Hypothesis Test!

Hypothesis testing for correlation

Solution 23

Student	English	Maths
Model mean	60	70
Model SD	5	10
1	63.90	70.13
2	55.24	65.99
3	58.80	80.18
4	59.65	57.16
5	66.44	72.76
6	59.53	79.82
7	57.43	71.48
8	58.33	60.56
9	67.43	69.56
10	63.11	87.13
Sample r	0.219	

One-tailed critical values: product moment (first five columns) and Spearman (last three).

n	0.10	0.05	0.025	0.01	0.005	0.05	0.025	0.01
4	.8000	.9000	.9500	.9800	.9900	1.0000	—	—
5	.6870	.8054	.8783	.9343	.9587	.9000	1.0000	1.0000
6	.6084	.7293	.8114	.8822	.9172	.8286	.8857	.9429
7	.5509	.6694	.7545	.8329	.8745	.7143	.7857	.8929
8	.5067	.6215	.7067	.7887	.8343	.6429	.7381	.8333
9	.4716	.5822	.6664	.7498	.7977	.6000	.7000	.7833
10	.4428	.5494	.6319	.7155	.7646	.5636	.6485	.7455
11	.4187	.5214	.6021	.6851	.7348	.5364	.6182	.7091
12	.3981	.4973	.5760	.6581	.7079	.5035	.5874	.6783
13	.3802	.4762	.5529	.6339	.6835	.4835	.5604	.6484
14	.3646	.4575	.5324	.6120	.6614	.4637	.5385	.6264

Test for positive correlation between English and Mathematics marks at the 10% significance level.

Null hypothesis $H_0 : \rho = 0$. There is no underlying correlation.

Alternative hypothesis

H1: ρ > 0. There is an underlying positive correlation.

Sample size $n = 10$.

Critical value

The 10% one-tailed critical value is 0.4428 (see table).

Conclusion

$0.219 < 0.4428$. Do not reject H_0. There is insufficient evidence of positive correlation between the marks.

The critical value is the minimum correlation required for rejection. Compare the values, then give the conclusion in context.

Two Tailed-test

Solution 24

In the previous example the alternative was positive correlation. A two-tailed test allows either direction.

A scientist takes 30 observations of the masses of two reactants and finds $r = -0.45$. Test her claim of no correlation at the 10% significance level. State your hypotheses.

One-tailed PMCC critical values

n	0.10	0.05	0.025	0.01	0.005
4	.8000	.9000	.9500	.9800	.9900
5	.6870	.8054	.8783	.9343	.9587
6	.6084	.7293	.8114	.8822	.9172
20	.2992	.3783	.4438	.5155	.5614
21	.2914	.3687	.4329	.5034	.5487
22	.2841	.3598	.4227	.4921	.5368
23	.2774	.3515	.4133	.4815	.5256
24	.2711	.3438	.4044	.4716	.5151
25	.2653	.3365	.3961	.4622	.5052
26	.2598	.3297	.3882	.4534	.4958
27	.2546	.3233	.3809	.4451	.4869
28	.2497	.3172	.3739	.4372	.4785
29	.2451	.3115	.3673	.4297	.4705
30	.2407	.3061	.3610	.4226	.4629
40	.2070	.2638	.3120	.3665	.4026
50	.1843	.2353	.2787	.3281	.3610
60	.1678	.2144	.2542	.2997	.3301

Null hypothesis $H_0 : \rho = 0$.

Alternative hypothesis $H_1 : \rho \neq 0$.

Sample size $n = 30$.

Tail probability $0.10/2 = 0.05$ per tail. Two-tailed: split the significance equally.

Critical values $r = \pm 0.3061$.

Conclusion

$-0.45 < -0.3061$, so reject H_0. At the 10% level there is evidence of correlation between the masses of the two reactants.

Test your understanding — Solution 25

[Textbook] The table from the large data set shows the daily maximum gust, x kn, and the daily maximum relative humidity, y%, in Leeming for a sample of eight days in May 2015.

x	31	28	38	37	18	17	21	29
y	99	94	87	80	80	89	84	86

a. Find the product moment correlation coefficient for this data.

b. Test, at the 10% level of significance, whether there is evidence of a positive correlation between daily maximum gust and daily maximum relative humidity. State your hypotheses clearly.

a $r = 0.1149$.

b

$H_0 : \rho = 0$, $H_1 : \rho > 0$. Sample size $n = 8$.

The 10% critical value is 0.5067. Since $0.1149 < 0.5067$, do not reject H_0.

There is insufficient evidence of positive correlation between maximum gust and maximum relative humidity.

Exercise 1C – Statistics Year 2 Pages 10-12 — Solution 26

3. (a) What does the product moment correlation coefficient measure? (1 mark)

Twelve students sat theoretical and practical Biology tests.

t	5	9	7	11	20	4	6	17	12	10	15	16
p	6	8	9	13	20	9	8	17	14	8	17	18

(b) Find the PMCC to 3 s.f. (2 marks)

(c) Test the claim that high theoretical marks tend to accompany high practical marks. Use the 5% level and state the hypotheses. (3 marks)

(d) Interpret 0.05 in this test. (1 mark)

a It measures the direction and strength of linear correlation between two variables.

b The calculator gives $r = 0.935$.

c

$H_0 : \rho = 0$, $H_1 : \rho > 0$; one tail, $\alpha = 0.05$.

$r = 0.935 > 0.4973$, so reject H_0. The marks show evidence of positive correlation. Students who score highly in the theoretical test tend to score highly in the practical test.

d

If there is no population correlation, the test has a 5% probability of rejecting the null hypothesis.

Exercise 1C – Statistics Year 2 Pages 10-12

Solution 27

4. The table shows test marks for eight students.

Student	*A*	*B*	*C*	*D*	*E*	*F*	*G*	*H*
English	25	18	32	27	21	35	28	30
Mathematics	16	11	20	17	15	26	32	20

(a) Calculate the PMCC. (1 mark)

(b) Test at the 5% significance level whether there is evidence of a linear relationship. State your hypotheses clearly. (3 marks)

a

$r = 0.68556\ldots \approx 0.686$ (3 s.f.). Use the calculator's correlation function.

b

$H_0 : \rho = 0$, $H_1 : \rho \neq 0$.

Either direction is possible. With $n = 8$, the two-tailed 5% critical value is 0.7067.

$|r| < 0.7067$, so do not reject H_0. There is insufficient evidence of a linear relationship between English and Mathematics marks.

Exercise 1C – Statistics Year 2 Pages 10-12

Solution 28

6. A researcher measures the amount of solvent in a solution and the reaction rate for 15 samples. She finds $r = -0.43$. Test her claim of negative correlation at the 5% level, stating your hypotheses. (3 marks)

6

$H_0 : \rho = 0$, $H_1 : \rho < 0$.

The critical value is -0.4409. Since $-0.43 > -0.4409$, do not reject H_0. There is insufficient evidence of negative correlation between solvent amount and reaction rate.

9. A weather centre employee believes humidity and visibility are negatively correlated. The sample is from Heathrow, August 1987.

Humidity (%)	92	93	91	82	91	100
Visibility (m)	2500	1500	2700	2900	2200	1000

© Crown Copyright Met Office

(a) Calculate the PMCC. (1 mark)

(b) Test the claim at the 1% level, stating your hypotheses. (3 marks)

9a $r = -0.846$ (3 s.f.).

9b

$H_0 : \rho = 0$, $H_1 : \rho < 0$.

The critical value is -0.8822. Since $-0.846 > -0.8822$, do not reject H_0. There is insufficient evidence of negative correlation between humidity and visibility.